Nicholas P. McKay

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Research Interests & Expertise

Dr. McKay researches climate dynamics on a range of spatial and temporal scales, ranging from regional to global, and decades to millennia. He field and lab work to develop new records of past climate, develops cyberinfrastructure to support paleoclimate research and uses artificial intelligence and machine learning techniques to integrate paleoclimate observations and numerical models. These research areas help us better understand how and why climate changed in the past, and how it will change in the future.

Appointments

2012-present — **Northern Arizona University**
School of Earth and Sustainability
Affiliate in the School of Informatics, Computing, and Cyber Systems

Professor - (2025 to current)
Associate Director, Institute Advancing Applications of Artificial Intelligence - (2026 to current)
Associate Professor - (2019 to 2025)

2021 — **GNS Science and Victoria University of Wellington, New Zealand**
Fulbright New Zealand Scholar

2012-present — **Northern Arizona University**
School of Earth and Sustainability

Assistant Professor - (2014 to 2019)
Assistant Research Professor - (2013 to 2014)
Postdoctoral Scholar - (2012 to 2013)

Education

2012 — **Doctor of Philosophy**, University of Arizona

Major: Geosciences
Minor: Global Change
Advisor: Dr. Jonathan T. Overpeck

2007 — **Master of Science**, Northern Arizona University, *with distinction*

Major: Geology
Advisor: Dr. Darrell S. Kaufman

2005 — **Bachelors of Science**, Northern Arizona University, *magna cum laude*

Majors: Geology & Environmental Science
Minor: Mathematics

Peer-reviewed Publications

**Denotes NAU student as lead author*

In Review

99. *Telles, F. T., S. Semken, **N. P. McKay**, and M. Nelson (2026a), Evaluating a place-based professional development workshop in geoscience for K-12 educators serving Indigenous communities in northern Arizona, *Journal of Geoscience Education*, in review
98. *Lathrop, E., S. Opfergelt, C. See, J. Mommer, N. Sandron, M. Thomas, C. Ebert, A. Kelley, **N. P. McKay**, and E. A. Schuur (2026), Permafrost thaw-driven loss of mineral-associated organic carbon, *Proceedings of the National Academy of Sciences*, in review
97. Kaufman, D., S. Hunt, J. Smith, J. Bright, C.-D. Hillenbrand, J. Klages, **N. P. McKay**, M. O'Regan, B. Risebrobakken, and G. West (2026), A calibration-based age calculator for high-latitude Quaternary foraminifera using amino acid racemization, *Scientific Data*, in review

In Revision

96. Khider, D., J. Emile-Geay, V. Ratnakar, **N. P. McKay**, J. P. Landers, Y. Gil, D. Garijo, D. Edge, C. C. Routson, T. Gondhalekar, and F. Zhu (2026a), LinkedEarth: a research ecosystem for open and reproducible paleoclimatology, *Paleoceanography and Paleoclimatology*, in revision, preprint doi:10.22541/essoar.15008010/v1
95. Mark, S. Z., B. V. Gaglioti, E. K. Thomas, D. H. Mann, G. A. Otiniano, C. L. Hancock, and **N. P. McKay** (2026), Deglacial constraints on the permafrost-carbon feedback, *AGU Advances*, in revision
94. *Bock, B., **N. P. McKay**, et al. (2026), Mapping the geographic, taxonomic, and socioeconomic biases of global fungal endophyte research, *New Phytologist*, in revision
93. Oswal, D., J. Pujara, **N. P. McKay**, D. Edge, and D. Khider (2026), PyleoTUPS, *Journal of Open Source Software*, in revision

2026

92. *Marshall, L. P., D. S. Kaufman, **N. P. McKay**, E. K. Thomas, H. Haflidason, M. Melles, and J. I. Svendsen (2026), Millennial-scale carbon-source changes revealed by sedimentary radiocarbon in Arctic lakes, *Quaternary Science Reviews*, *390*, 110,121
91. *Telles, F. T., **N. P. McKay**, K. E. Davis, and S. Arcusa (2026b), Spanish colonization triggered 17th-century landscape dust regime shift in the Southwestern United States, *Anthropocene*, *54*, 100,553
90. Larocca, L. J., D. S. Kaufman, M. P. Erb, T. M. Schultz, **N. P. McKay**, J. P. Brooks, and J. L. Carrivick (2026), Glacier equilibrium-line altitude change across Alaska and adjacent Canada indicates a cold, dry Little Ice Age and weaker Aleutian Low, *Geophysical Research Letters*, *53*(4), e2025GL119,840
89. *Allen, H. T., D. S. Kaufman, **N. P. McKay**, E. Schiefer, L. J. Larocca, B. J. L. Jensen, M. S. M. Bolton, K. R. North, N. Praet, M. Van Daele, M. De Batist, and M. Żarczyński (2026), Holocene glacier fluctuations at Eklutna Lake, Chugach Mountains, south-central Alaska, *Quaternary Science Reviews*, *390*, 110,107

2025

88. Kaufman, D. S., J. Bright, C. Ebert, K. Lindberg, L. Marshall, **N. P. McKay**, E. Schuur, and E. Thomas (2025), Radiocarbon dating of lake sediment using low-temperature combustion, *Radiocarbon*, pp. 1–22
87. Khider, D., J. Emile-Geay, D. Edge, and **N. P. McKay** (2025), Empowering Learners-Teaching Reproducible Research with Open-Source Tools, pp. 303–310

2025

86. Praet, N., M. Van Daele, K. Wils, P. J. Haeussler, R. C. Witter, **N. P. McKay**, B. J. Jensen, J. Moernaut, and M. De Batist (2025), Refining the earthquake history of south-central Alaska through lake records, *Earth-Science Reviews*, p. 105249

85. Paine, A. R., J. Frieling, T. M. Shanahan, T. A. Mather, **N. P. McKay**, S. A. Robinson, D. M. Pyle, I. M. Fendley, R. Kiely, and W. D. Gosling (2025), Evidence for millennial-scale interactions between Hg cycling and hydroclimate from Lake Bosumtwi, Ghana, *Climate of the Past*, *21*(4), 817–839

84. Holtzman, H., E. K. Thomas, M. Erb, L. Marshall, I. S. Castañeda, D. Kaufman, **N. P. McKay**, and M. Melles (2025), Early Holocene atmospheric circulation changes over northern Europe based on isotopic and biomarker evidence from Kola Peninsula, *Paleoceanography and Paleoclimatology*, *40*(3), e2024PA005,076

83. *Marshall, L. P., D. S. Kaufman, N. H. Bigelow, M. S. Bolton, E. C. Cohen, B. M. Damon, B. P. Finney, B. J. Jensen, **N. P. McKay**, M. J. Medina, et al. (2025), A 15,800 year record of productivity, carbon accumulation and environmental change at Eight Mile Lake and its catchment, central Alaska, *Arctic, Antarctic, and Alpine Research*, *57*(1), 2523,083

2024

82. *Hancock, C. L., M. P. Erb, **N. P. McKay**, S. G. Dee, and R. F. Ivanovic (2024), A global Data Assimilation of Moisture Patterns from 21 000–0 BP (DAMP-21ka) using lake level proxy records, *Climate of the Past*, *20*(12), 2663–2684

81. Arzey, A. K., H. V. McGregor, T. R. Clark, J. M. Webster, S. E. Lewis, J. Mallela, **N. P. McKay**, H. W. Fahey, S. Chakraborty, T. B. Razak, and M. J. Fischer (2024), Coral skeletal proxy records database for the Great Barrier Reef, Australia, *Earth System Science Data*, *16*(10), 4869–4930

80. **McKay, N. P.**, D. S. Kaufman, S. H. Arcusa, H. Kolus, D. Edge, M. P. Erb, C. Hancock, C. C. Routson, M. Żarczyński, L. P. Marshall, G. Roberts, and F. Telles (2024a), The 4.2 ka event is not remarkable in the context of Holocene climate variability, *Nature Communications*, *15*(1), 6555

79. Wheeler, C., J. K. Pearman, J. D. Howarth, M. J. Vandergoes, K. Holt, S. A. Trewick, X. Li, L. Thompson, G. Thomson-Laing, M. Picard, C. Moy, **N. P. McKay**, A. Moody, C. Shepherd, V. van den Bos, K. Steiner, and S. A. Woodothers (2024), A paleoecological investigation of recent cyanobacterial blooms and their drivers in two contrasting lakes, *Harmful Algae*, *131*, 102,563

78. Thomson-Laing, G., J. D. Howarth, J. Atalah, M. J. Vandergoes, X. Li, J. K. Pearman, S. Fitzsimons, C. Moy, A. Moody, C. Shepherd, **N. P. McKay**, and S. Wood (2024), Sedimentary ancient DNA reveals the impact of anthropogenic land use disturbance and ecological shifts on fish community structure in small lowland lake, *Science of The Total Environment*, p. 171266

77. Larocca, L. J., J. M. Lea, M. P. Erb, **N. P. McKay**, M. Phillips, K. A. Lamantia, and D. S. Kaufman (2024), Arctic glacier snowline altitudes rise 150 m over the last 4 decades, *The Cryosphere*, *18*(8), 3591–3611, doi: 10.5194/tc-18-3591-2024

2023

76. Konecky, B. L., **N. P. McKay**, G. M. Falster, S. L. Stevenson, M. J. Fischer, A. R. Atwood, D. M. Thompson, M. D. Jones, J. J. Tyler, K. L. DeLong, et al. (2023), Globally coherent water cycle response to temperature change during the past two millennia, *Nature Geoscience*, *16*(11), 997–1004

75. *Marshall, L. P., D. S. Kaufman, R. S. Anderson, **N. P. McKay**, and E. A. Schuur (2023), Organic-Matter Accumulation and Degradation in Holocene Permafrost Deposits Along a Central Alaska Hillslope, *Journal of Geophysical Research: Biogeosciences*, *128*(9), e2022JG007,290

2023

74. Zhu, F., J. Emile-Geay, K. J. Anchukaitis, **N. P. McKay**, S. Stevenson, and Z. Meng (2023), A pseudoproxy emulation of the PAGES 2k database using a hierarchy of proxy system models, *Scientific Data*, *10*(1), 624
73. Cluett, A. A., E. K. Thomas, **N. P. McKay**, O. C. Cowling, I. S. Castañeda, and C. Morrill (2023), Lake Dynamics Modulate the Air Temperature Variability Recorded by Sedimentary Aquatic Biomarkers: A Holocene Case Study From Western Greenland, *Journal of Geophysical Research: Biogeosciences*, *128*(7), e2022JG007,106, doi: https://doi.org/10.1029/2022JG007106
72. *Thurston, L. L., E. Schiefer, **N. P. McKay**, and D. S. Kaufman (2023), Suspended sediment deposition and sediment yields at Lake Peters, northeast Brooks Range, Alaska: Fluvial-and lake-based approaches, *Earth Surface Processes and Landforms*, *48*(13), 2521–2535
71. Thomas, E. K., A. A. Cluett, M. P. Erb, **N. P. McKay**, J. P. Briner, I. S. Castañeda, M. C. Corcoran, O. C. Cowling, D. B. Gorbey, K. R. Lindberg, et al. (2023), Early Holocene Laurentide Ice Sheet Retreat Influenced Summer Atmospheric Circulation in Baffin Bay, *Geophysical Research Letters*, *50*(13), e2023GL103,428
70. *Hancock, C. L., **N. P. McKay**, M. P. Erb, D. S. Kaufman, C. R. Routson, R. F. Ivanovic, L. J. Gregoire, and P. Valdes (2023), Global Synthesis of Regional Holocene Hydroclimate Variability Using Proxy and Model Data, *Paleoceanography and Paleoclimatology*, p. e2022PA004597
69. Walter, R. M., H. R. Sayani, T. Felis, K. M. Cobb, N. J. Abram, A. K. Arzey, A. R. Atwood, L. D. Brenner, É. P. Dassié, K. L. DeLong, B. Ellis, M. J. Fischer, N. F. Goodkin, J. A. Hargreaves, H. K. K Halimeda Kilbourne, **N. P. McKay**, S. A. Murty, R. D. Ramos, E. V. Reed, D. Samanta, S. C. Sanchez, and J. Zinke (2023), The CoralHydro2k Database: a global, actively curated compilation of coral $\delta^{18}O$ and Sr/Ca proxy records of tropical ocean hydrology and temperature for the Common Era, *Earth System Science Data*, *15*, 2081—2116, doi: 10.5194/essd-15-2081-2023

2022

68. Erb, M. P., **N. P. McKay**, N. Steiger, S. Dee, C. Hancock, R. F. Ivanovic, L. J. Gregoire, and P. Valdes (2022), Reconstructing Holocene temperatures in time and space using paleoclimate data assimilation, *Climate of the Past*, *18*(12), 2599–2629
67. Routson, C. C., M. P. Erb, and **N. P. McKay** (2022), High latitude modulation of the Holocene North American monsoon, *Geophysical Research Letters*, *49*(16), e2022GL099,772
66. **McKay, N. P.**, J. Emile-Geay, and D. Khider (2022a), Database interoperability, uncertainty quantification and reproducible workflows in the paleogeosciences (Version ec2022), *Earthcube Annual Meeting 2022 - Computational notebooks*, doi: 10.5281/zenodo.6780665
65. Thompson, D., J. Conroy, B. Konecky, S. Stevenson, K. DeLong, **N. P. McKay**, E. Reed, L. Jonkers, and M. Carré (2022), Identifying hydro-sensitive coral δ^{18}O records for improved high-resolution temperature and salinity reconstructions, *Geophysical Research Letters*, p. e2021GL096153
64. Kaufman, D. S., and **N. P. McKay** (2022), Past and future warming–direct comparison on multi-century timescales, *Climate of the Past*, *18*(4), 911–917
63. *Arcusa, S. H., **N. P. McKay**, C. Wiman, S. Patterson, S. E. Munoz, and M. A. Aquino-López (2022), A Bayesian approach to integrating radiometric dating and varve measurements in intermittently indistinct sediment, *Quaternary Geochronology*, *4*(1), 409–433
62. Manety, S., D. Khider, C. Heiser, **N. P. McKay**, J. Emile-Geay, and C. Routson (2022), Paleo-Rec: A sequential recommender system for the annotation of paleoclimate datasets, *Environmental Data Science*, *1*
61. *Broadman, E., D. S. Kaufman, R. S. Anderson, S. Bogle, M. Ford, D. Fortin, A. C. Henderson, J. H. Lacey, M. J. Leng, **N. P. McKay**, et al. (2022), Reconstructing postglacial hydrologic and environmental change in the eastern Kenai Peninsula lowlands using proxy data and mass balance modeling, *Quaternary Research*, *107*, 1–26

2022

60. Pearman, J. K., S. A. Wood, M. J. Vandergoes, J. Atalah, S. Waters, J. Adamson, G. Thomson-Laing, L. Thompson, J. D. Howarth, D. P. Hamilton, X. Pochon, L. Biessy, K. A. Brasell, J. Dahl, R. Ellison, S. J. Fitzsimons, H. Gard, T. Gerrard, R. Gregersen, M. Holloway, X. Li, D. J. Kelly, R. Martin, K. McFarlane, **N. P. McKay**, A. Moody, C. M. Moy, S. Naeher, R. Newnham, R. Parai, M. Picard, J. Puddick, A. B. Rees, L. Reyes, M. Schallenberg, C. Shepherd, J. Short, K. S. Simon, K. Steiner, C. Å unde, M. Terezow, and J. Tibby (2022), A bacterial index to estimate lake trophic level: National scale validation, *Science of The Total Environment*, *812*, 152,385, doi: https://doi.org/10.1016/j.scitotenv.2021.152385

2021

59. Schiefer, E., J. Geck, J. S. Ostman, **N. P. McKay**, N. Praet, M. G. Loso, and D. S. Kaufman (2021), Fluvial suspended sediment transfer and lacustrine sedimentation of recent flood turbidites in proglacial Eklutna Lake, western Chugach Mountains, Alaska, *Hydrological Processes*, *35*(10), e14,375
58. Sun, C., T. Shanahan, P. DiNezio, **N. P. McKay**, and P. Roy (2021), Spring warming drives past and future shifts in Great Plains storm intensity, *Nature Geoscience*, *14*(12), 912–917
57. *Thomas, J. E., D. S. Kaufman, N. Praet, **N. P. McKay**, M. V. Daele, B. J. Jensen, and M. D. Batist (2021), A 2300-year record of glacier fluctuations at Skilak and Eklutna Lakes, south-central Alaska, *Quaternary Science Reviews*, *272*, 107,215
56. Routson, C. C., D. S. Kaufman, **N. P. McKay**, M. P. Erb, S. H. Arcusa, K. J. Brown, M. E. Kirby, J. P. Marsicek, R. S. Anderson, G. Jiménez-Moreno, et al. (2021), A multiproxy database of western North American Holocene paleoclimate records, *Earth System Science Data*, *13*(4), 1613–1632
55. **McKay, N. P.**, J. Emile-Geay, and D. Khider (2021a), geoChronR–an R package to model, analyze, and visualize age-uncertain data, *Quaternary Geochronology*, *3*(1), 149–169

2020

54. Turney, C. S., R. T. Jones, **N. P. McKay**, E. Van Sebille, Z. A. Thomas, C.-D. Hillenbrand, and C. J. Fogwill (2020a), A global mean sea surface temperature dataset for the Last Interglacial (129–116 ka) and contribution of thermal expansion to sea level change, *Earth System Science Data*, *12*(4), 3341–3356
53. *Arcusa, S. H., **N. P. McKay**, C. M. Carillo, and T. R. Ault (2020b), Dust-Drought Nexus in the southwestern United States: A Proxy-Model Comparison Approach, *Paleoceanography and Paleoclimatology*, *35*(12), e2020PA004,046
52. Konecky, B. L., **N. P. McKay**, O. V. Churakova (Sidorova), L. Comas-Bru, E. P. Dassié, K. L. DeLong, G. M. Falster, M. J. Fischer, M. D. Jones, L. Jonkers, D. S. Kaufman, G. Leduc, S. R. Managave, B. Martrat, T. Opel, A. J. Orsi, J. W. Partin, H. R. Sayani, E. K. Thomas, D. M. Thompson, J. J. Tyler, N. J. Abram, A. R. Atwood, J. L. Conroy, Z. Kern, T. J. Porter, S. L. Stevenson, and L. von Gunten (2020), The Iso2k Database: A global compilation of paleo-δ^{18}O and δ^{2}H records to aid understanding of Common Era climate, *Earth System Science Data*, *12*(3), 2261–2288, doi: 10.5194/essd-2020-5
51. Kjellman, S. E., A. Schomacker, E. K. Thomas, L. Håkansson, S. Duboscq, A. A. Cluett, W. R. Farnsworth, L. Allaart, O. C. Cowling, **N. P. McKay**, et al. (2020), Holocene precipitation seasonality in northern Svalbard: Influence of sea ice and regional ocean surface conditions, *Quaternary Science Reviews*, *240*, 106,388
50. Kaufman, D., **N. P. McKay**, C. Routson, M. Erb, C. Dätwyler, P. Sommer, O. Heiri, and B. Davis (2020a), Holocene global mean surface temperature: A multi-method reconstruction approach, *Scientific data*, *7*(1), 1–13

2020

49. Thomas, Z. A., R. T. Jones, C. S. Turney, N. Golledge, C. Fogwill, C. J. Bradshaw, L. Menviel, **N. P. McKay**, M. Bird, J. Palmer, et al. (2020), Tipping elements and amplified polar warming during the Last Interglacial, *Quaternary Science Reviews*, *233*, 106,222
48. Turney, C. S., C. J. Fogwill, N. R. Golledge, **N. P. McKay**, E. van Sebille, R. T. Jones, D. Etheridge, M. Rubino, D. P. Thornton, S. M. Davies, et al. (2020b), Early Last Interglacial ocean warming drove substantial ice mass loss from Antarctica, *Proceedings of the National Academy of Sciences*, *117*(8), 3996–4006
47. *Thurston, L. L., E. Schiefer, **N. P. McKay**, and D. S. Kaufman (2020), Modelling suspended sediment discharge in a glaciated Arctic catchment–Lake Peters, Northeast Brooks Range, Alaska, *Hydrological Processes*, *34*(19), 3910–3927
46. Brierley, C. M., A. Zhao, S. P. Harrison, P. Braconnot, C. J. R. Williams, D. J. R. Thornalley, X. Shi, J.-Y. Peterschmitt, R. Ohgaito, D. S. Kaufman, M. Kageyama, J. C. Hargreaves, M. P. Erb, J. Emile-Geay, R. D'Agostino, D. Chandan, M. Carré, P. J. Bartlein, W. Zheng, Z. Zhang, Q. Zhang, H. Yang, E. M. Volodin, R. A. Tomas, C. Routson, W. R. Peltier, B. Otto-Bliesner, P. A. Morozova, **N. P. McKay**, G. Lohmann, A. N. Legrande, C. Guo, J. Cao, E. Brady, J. D. Annan, and A. Abe-Ouchi (2020), Large-scale features and evaluation of the PMIP4-CMIP6 *midHolocene* simulations, *Climate of the Past*, *16*(5), 1847–1872, doi: 10.5194/cp-16-1847-2020
45. Jonkers, L., O. Cartapanis, M. Langner, **N. P. McKay**, S. Mulitza, A. Strack, and M. Kucera (2020), Integrating palaeoclimate time series with rich metadata for uncertainty modelling: strategy and documentation of the PALMOD 130k marine palaeoclimate data synthesis, *Earth System Science Data*, *12*(2), 1053–1081, doi: 10.5194/essd-12-1053-2020
44. Kaufman, D., **N. P. McKay**, C. Routson, M. Erb, B. Davis, O. Heiri, S. Jaccard, J. Tierney, C. Dätwyler, et al. (2020b), A global database of Holocene paleotemperature records, *Scientific data*, *7*(1), 183
43. *Arcusa, S., **N. P. McKay**, C. Routson, and S. Munoz (2020a), Dust-drought interactions over the last 15,000 years: a network of lake sediment records from the San Juan Mountains, Colorado, *The Holocene*, *30*(4), 559–574

2019

42. *Broadman, E., L. L. Thurston, E. Schiefer, D. Fortin, **N. P. McKay**, J. Geck, M. G. Loso, M. Nolan, S. H. Arcusa, C. W. Benson, R. A. Ellerbroek, M. P. Erb, C. C. Routson, C. Wiman, A. J. Wong, and D. S. Kaufman (2019), An Arctic watershed observatory at Lake Peters, Alaska: weather–glacier–river–lake system data for 2015–2018, *Earth System Science Data*, *11*(4), 1957–1970, doi: 10.5194/essd-11-1957-2019
41. Khider, D., J. Emile-Geay, **N. P. McKay**, Y. Gil, D. Garijo, V. Ratnakar, M. Alonso-Garcia, S. Bertrand, O. Bothe, P. Brewer, et al. (2019), PaCTS 1.0: a crowdsourced reporting standard for paleoclimate data, *Paleoceanography and paleoclimatology*, *34*(10), 1570–1596
40. *Benson, C. W., D. S. Kaufman, **N. P. McKay**, E. Schiefer, and D. Fortin (2019), A 16,000-yr-long sedimentary sequence from Lakes Peters and Schrader (Neruokpuk Lakes), northeastern Brooks Range, Alaska, *Quaternary Research*, pp. 1–17
39. Routson, C. C., S. H. Arcusa, **N. P. McKay**, and J. T. Overpeck (2019a), A 4,500-Year-Long Record of Southern Rocky Mountain Dust Deposition, *Geophysical Research Letters*, *46*(14), 8281–8288
38. PAGES 2k Consortium (2019), Consistent multi-decadal variability in global temperature reconstructions and simulations over the Common Era, *Nature geoscience*, *12*(8), 643–649
37. Fortin, D., N. Praet, **N. P. McKay**, D. S. Kaufman, B. J. Jensen, P. J. Haeussler, C. Buchanan, and M. De Batist (2019), New approach to assessing age uncertainties–The 2300-year varve chronology from Eklutna Lake, Alaska (USA), *Quaternary Science Reviews*, *203*, 90–101

2019

36. Routson, C. C., **N. P. McKay**, D. S. Kaufman, M. P. Erb, H. Goosse, B. N. Shuman, J. R. Rodysill, and T. Ault (2019b), Mid-latitude net precipitation decreased with Arctic warming during the Holocene, *Nature*, *568*(7750), 83

35. Zhu, F., J. Emile-Geay, **N. P. McKay**, G. J. Hakim, D. Khider, T. R. Ault, E. J. Steig, S. Dee, and J. W. Kirchner (2019), Climate models can correctly simulate the continuum of global-average temperature variability, *Proceedings of the National Academy of Sciences*, *116*(18), 8728–8733

34. *Kolus, H. R., D. N. Huntzinger, C. R. Schwalm, J. B. Fisher, **N. P. McKay**, Y. Fang, A. M. Michalak, K. Schaefer, Y. Wei, B. Poulter, J. Mao, N. C. Parazoo, and X. Shi (2019a), Land carbon models underestimate the severity and duration of droughtâ ™s impact on plant productivity, *Scientific reports*, *9*(1), 2758

2018

33. **McKay, N.P.**, D. S. Kaufman, C. C. Routson, M. P. Erb, and P. D. Zander (2018a), The onset and rate of Holocene Neoglacial cooling in the Arctic, *Geophysical Research Letters*, *45*(22), 12–487

32. Thomas, E., I. Castaneda, **N. P. McKay**, , J. Briner, J. Salacup, K. Nguyen, and A. Schweinsberg (2018), Arctic hydroclimate intensification coincident with hemispheric warming 8,000 years ago, *Geophysical Research Letters*, *45*(19), 10–637

31. Levy, R. H., G. B. Dunbar, M. J. Vandergoes, J. D. Howarth, T. Kingan, A. R. Pyne, G. Brotherston, M. Clarke, B. Dagg, M. Hill, E. Kenton, S. Little, D. Mandeno, C. Moy, P. Muldoon, P. Doyle, C. Raines, P. Rutland, D. Strong, M. Terezow, L. Cochrane, R. Cossu, S. Fitzsimons, F. Florindo, A. L. Forrest, A. R. Gorman, D. S. Kaufman, M. K. Lee, X. Li, P. Lurcock, **N. P. McKay**, F. Nelson, J. Purdie, H. A. Roop, S. G. Schladow, A. Sood, P. Upton, S. L. Walker, and G. S. Wilson (2018), A high-resolution climate record spanning the past 17,000 years recovered from Lake Ohau, South Island, New Zealand, *Scientific Drilling*, doi: 10.5194/sd-6-1-2018

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Other Publications

2026

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2024

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2023

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2022

9. Kaufman, D., J. Hunter, and **N. P. McKay** (2022), PAGES 2k data portal and the LiPDverse, *PAGES Magazine*, *30*(2), 131

2018

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7. **McKay, N. P.**, and J. Emile-Geay (2018), Linked PaleoData: A resource for open, reproducible, and efficient paleoclimatology, *PAGES News*, *26*, 71

6. Hakim, G., S. Dee, J. Emile-Geay, **N. P. McKay**, and K. Rehfeld (2018), Accelerating Progress in Proxy–Model Synthesis Using Open Standards, *PAGES News*, *26*, 73

2016

5. Emile-Geay, J., and **N. P. McKay** (2016), Paleoclimate data standards, *PAGES News*, *24*(1), 47

2014

4. **McKay, N.P.** (2014a), A novel multiproxy approach: The PAGES North America 2k working group, *PAGES News*, *22*(2), 100
3. Anchukaitis, K., and **N. P. McKay** (2014), PAGES2k: Advances in climate field reconstructions, *PAGES News*, *22*(2), 98

2013

2. Kaufman, D., **N. P. McKay**, T. Kiefer, and L. von Gunten (2013), A Regional View of Climate Change, *Global Change Magazine*, *81*, 18–23
1. von Gunten, L., D. Anderson, B. Chase, M. Curran, J. Gergis, E. Gille, W. Gross, S. Hanhijärvi, D. Kaufman, T. Kiefer, **N. P. McKay**, I. Mundo, R. Neukom, M. Sano, A. Shah, J. Tyler, A. Viau, S. Wagner, E. Wahl, and D. Willard (2013), The backbone of PAGES 2k: data management and archiving, *PAGES News*, *21*(2), 87

Presentations

Only presentations given by myself, or students for whom I was lead advisor, are listed. Other coauthored presentations omitted

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2026

82. **McKay, N. P.**, D. Edge, M. P. Erb, J. Emile-Geay, D. Khider, and T. Gondhalekar (2026), PReSto: An Open Platform for Reproducible, Custom Paleoclimate Data Assimilation Reconstructions, *AGU Annual Meeting, San Francisco, CA*, [Oral]
81. **McKay, N. P.**, and T. Curry (2026), Building an AI Agent to Support Grant and Publication Writing and Submission (Copilot), *SHERC Workshop, Northern Arizona University, Flagstaff, AZ*, [Workshop]
80. Khider, D., **N. P. McKay**, D. Edge, and J. Pujara (2026b), TUPS: Table Understanding for Paleoclimate Studies, *2026 NSF CSSI/CyberTraining/SCIPE PI Meeting, Rockville, MD*, doi: 10.6084/m9.figshare.29622500.v2, [Poster]

2025

79. *Medlin, D., and **N. P. McKay** (2025), Extending Drought Risk Assessment Through the Past Millennium: Machine Learning Integration of Paleoclimate PDSI and Modern USDM Data. AGU Fall Meeting, New Orleans, LA, [Oral]
78. Khider, D., and **N. P. McKay** (2025), TUPS: Table Understanding for Paleoclimate Studies, *2025 NSF CSSI/CyberTraining/SCIPE PI Meeting, Denver, CO*, doi: 10.6084/m9.figshare.29622500.v1, [Poster]
77. **McKay, N. P.** (2025), Paleoenvironmental Records of Extreme Events: Data curation, services and provenancing, *National Academies' consensus study on Functions and Criteria for a New Center for Paleoenvironmental Records of Extreme Events*, [Invited Talk]

2024

76. **McKay, N. P.**, D. S. Kaufman, S. Arcusa, D. Edge, M. P. Erb, C. Hancock, H. Kolus, C. C. Routson, M. Żarczyński, L. Marshall, G. Roberts, and F. Telles (2024b), The 4.2 ka event is not remarkable in the context of century-scale Holocene climate variability, *Seminar, University of California Santa Barbara*, [Invited Seminar]

2023

75. **McKay, N. P.**, D. Edge, M. Erb, D. Khider, J. Emile-Geay, and C. Routson (2023a), A Paleoclimate Reconstruction Storehouse (PReSto) to integrate paleoclimate data, reconstruction workflows and visualization, *Building Upon the EarthCube Community Workshop, Marina del Rey, California*, [Poster]
74. *Telles, F. (2023a), Columbine Lake discovers dust caused by colonization, *Religious Origins of White Supremacy, Syracuse University, New York*
73. *Telles, F. (2023b), The nexus between droughts, pluvials, and land use on dust emissions in the Colorado Plateau, *Ecological Society of America, Portland, Oregon*
72. **McKay, N. P.**, D. Kaufman, S. Arcusa, and H. Kolus (2023b), Was the 4.2 ka event unusual in context of global Holocene climate variability?, *EGU Spring Meeting, Vienna, Austria*, [Talk]
71. *Telles, F., and **N. P. McKay** (2023), The impacts of droughts, pluvials, and land use decisions on dust accumulation at Columbine Lake, Colorado, *International Conference on Aeolian Research, Las Cruces, New Mexico*
70. **McKay, N. P.** (2023), LiPD-Neotoma Integrations, *Neotoma All Hands Meeting*, [Invited Talk]
69. **McKay, N. P.**, D. S. Kaufman, S. Arcusa, D. Edge, M. P. Erb, C. Hancock, H. Kolus, C. C. Routson, M. Żarczyński, L. Marshall, G. Roberts, and F. Telles (2023c), The 4.2 ka event is not remarkable in the context of century-scale Holocene climate variability, *AGU Fall Meeting, San Francisco, CA*, [Poster]

2022

68. *Telles, F., **N. P. McKay**, and S. Arcusa (2022), How are dust and drought related over the past 3,000 years at Columbine Lake, Colorado?, *Society for Advancing Chicanos/Hispanic & Native Americans in Science, San Juan, Puerto Rico*
67. **McKay, N. P.** (2022), Abrupt Change in Climate And Ecosystems: Data and e-infrastructure, *Sustainability, Research and Innovation Congress, Praetoria, South Africa*, [Invited talk]
66. *Hancock, C., **N. P. McKay**, M. P. Erb, R. F. Ivanovic, L. J. Gregoire, and P. J. Valdes (2022), Global Synthesis of Regional Holocene Hydroclimate Variability Using Proxy and Model Data, *AGU Fall Meeting 2022*
65. **McKay, N. P.**, J. Emile-Geay, and D. Khider (2022b), Database interoperability, uncertainty quantification and reproducible workflows in the paleogeosciences (Version ec2022), *Earthcube Annual Meeting 2022*, [Talk]
64. **McKay, N. P.**, C. Routson, and M. Erb (2022c), High latitude modulation of the Holocene North American monsoon, *Biennial Meeting of the American Quaternary Association*, [Talk]

2021

63. **McKay, N. P.**, M. Vandergoes, and S. Wood (2021b), The Lakes380 project: Using sediments as indicators of current and historic lake health on a national scale in New Zealand, *AGU Fall Meeting*, *2021*, PP24A–02
62. *Hancock, C., **N. P. McKay**, M. Erb, L. Gregoire, R. Ivanovic, and P. Valdes (2021), Holocene Hydroclimate: Multi Millennial-Scale Trends in Proxies and Models, *AGU Fall Meeting*, *2021*, PP55A–0634

2021

61. **McKay, N. P.** (2021a), 2k Network Data Stewardship, *PAGES 2k Network Seminar Series*, [Invited Seminar]
60. **McKay, N. P.** (2021b), The mystery of Holocene temperature evolution, *Cawthron Institute, Nelson, New Zealand*, [Invited Seminar]
59. **McKay, N. P.** (2021c), The mystery of Holocene temperature evolution, *Victoria University of Wellington, Wellington, New Zealand*, [Invited Seminar]
58. **McKay, N. P.** (2021d), The mystery of Holocene temperature evolution, *GNS Science, Lower Hutt, New Zealand*, [Invited Seminar]
57. **McKay, N. P.** (2021e), The mystery of Holocene temperature evolution, *University of Otago, Dunedin, New Zealand*, [Invited Seminar]
56. **McKay, N. P.** (2021f), New tools for quantifying and propagating age uncertainty in varve records, *PAGES Varve Working Group Seminar Series*, [Invited Seminar]

2020

55. *Born, F., S. *Arcusa, **N. P. McKay**, and R. S. Anderson (2020), Regional Wildfire Inferred From Micro-Charcoal Over Past 15,000 years in Southern Colorado USA., in *AGU Fall Meeting*, vol. 2020, pp. PP047–0005
54. *Arcusa, S., and **N. P. McKay** (2020), Constraining the role of age uncertainty in event relationships in data-model comparison, in *AGU Fall Meeting*, vol. 2020, pp. PP043–02

2019

53. **McKay, N.P.** (2019a), A new global multiproxy Holocene temperature database: a resource for model comparison and evaluation, *AGU Fall Meeting*, *2019*, PP31A–07, [Talk]
52. *Wiman, C., **N. P. McKay**, C. Routson, S. Arcusa, and S. E. Munoz (2019), Late Holocene spring snowfall and hydroclimate variability inferred from varved sediments, Columbine Lake, Colorado, *AGU Fall Meeting*, *2019*, PP42B–04, [Talk]
51. *Kolus, H. R., **N. P. McKay**, D. S. Kaufman, and C. Routson (2019b), Is the 4.2 ka event remarkable in the context of Holocene centennial-scale climate variability?, *AGU Fall Meeting*, *2019*, GC11G–1127, [Poster]
50. **McKay, N.P.** (2019b), Climate change and risk, *Fall Meeting of The Casualty Actuaries of the Desert States (CADS), Tempe, Arizona*, [Invited Talk]
49. *Chaffeur, J., S. Arcusa, and **N. P. McKay** (2019), Examining environmental impact from mining using multiple proxies on a sediment core from Clear Lake, Colorado, USA, *GSA Annual Meeting, Phoenix, Arizona*, [Poster]
48. **McKay, N.P.** (2019c), Climate change and risk, *Arizona Capture Insurance Association Meeting, Phoenix, Arizona*, [Invited Talk]
47. **McKay, N.P.** (2019d), Linked PaleoData and the Climate12k database, *Climate12k workshop, Ste Croix, Switzerland*, [Invited Talk]

2018

46. *Arcusa, S., C. Routson, and **N. P. McKay** (2018a), Temporal variability of dust deposition from Holocene lake sediment records in the San Juan Mountains, CO., *Fall meeting of the American Geophysical Union, Washington, D.C*, [Poster]
45. *Broadman, E., D. S. Kaufman, A. C. Henderson, R. S. Anderson, E. Berg, M. J. Leng, and **N. P. McKay** (2018), Holocene hydroclimatic change in south-central Alaska inferred from $\delta^{18}O_{diatom}$ at Sunken Island Lake, Kenai Peninsula lowlands, *Fall meeting of the American Geophysical Union, Washington, D.C*, [Poster]

2018

44. *Kolus, H., **N. P. McKay**, and D. Kaufman (2018), How prominent was the 8.2 ka event? A global Holocene perspective, *Fall meeting of the American Geophysical Union, Washington, D.C*, [Poster]
43. **McKay, N.P.**, J. Emile-Geay, and D. Khider (2018b), Scientific workflows, reproducibility and uncertainty quantification in the paleogeosciences, *Fall meeting of the American Geophysical Union, Washington, D.C*, [Invited Talk]
42. *Wiman, C., **N. P. McKay**, C. Routson, R. S. Anderson, and S. Arcusa (2018a), Reconstructing Late Holocene Productivity and Climate in Columbine Lake, Colorado, *Fall meeting of the American Geophysical Union, Washington, D.C*, [Poster]
41. **McKay, N.P.** (2018), The Linked Paleo Data framework (LiPD) - introduction and open discussion , *1st PAGES Webinar - Wednesday 17 October 2018*, [Invited Talk]
40. *Wiman, C., **N. P. McKay**, R. S. Anderson, and C. Routson (2018b), Reconstructing Late Holocene productivity and climate at Columbine Lake, CO through annually laminated sediments, *Rocky Mountain Regional meeting of the Geological Society of America, Flagstaff, AZ*, [Poster]
39. *Arcusa, S., C. Routson, and **N. P. McKay** (2018b), Spatial and temporal variability of dust deposition in the San Juan Mountains, CO, *Dust Impact on Climate and the Environment, Las Cruces, Chile*, [Poster]

2017

38. *Arcusa, S., C. Routson, and **N. P. McKay** (2017), Spatial and temporal variability of dust deposition in the San Juan Mountains, CO: a network of Late Holocene lake sediment records, *Fall meeting of the American Geophysical Union, New Orleans, LA*, [Talk]
37. *Arcusa, S., and **N. P. McKay** (2017), 14,000 year lake sediment dust record from the San Juan Mountains, CO, *Colorado Plateau Biennial Conference, Flagstaff, AZ*, [Talk]
36. **McKay, N.P.**, D. Khider, J. Emile-Geay, D. Garijo, V. Ratnakar, and Y. Gil (2017), Supporting paleoclimate research with Semantic Web technologies and community Curation, *Earthcube All Hands Meeting 2017, Seattle WA*, [Poster]
35. **McKay, N.P.** (2017), Linked PaleoData: What is it and what can it do for you?, *PAGES 5th Open Science Meeting, Zaragoza, Spain*, [Talk]
34. **McKay, N.P.**, and the PAGES 2k Consortium (2017), Spatial covariability of temperature and hydroclimate as a function of timescale during the Common Era, *AGU Fall Meeting 2017, New Orleans, Louisiana*, [Talk]

2016

33. **McKay, N.P.** (2016a), On the PAGES2k project, paleoclimate cyberinfrastructure and integrating paleoclimatology and paleoseismology, *Southern California Earthquake Center, SoSAFE Workshop, Palm Springs, CA*, [Invited Talk]
32. **McKay, N.P.** (2016b), A global multiproxy database for temperature reconstructions of the common era, *American Quaternary Association, 24th Biennial Meeting, Santa Fe, NM*, [Invited Talk]
31. **McKay, N.P.** (2016c), The structure of Linked Paleo Data, *PaleoData Standards Workshop, Boulder, CO*, [Invited Talk]
30. **McKay, N.P.**, D. S. Kaufman, P. D. Zander, and C. C. Routson (2016), The timing and amplitude of Holocene Neoglaciation in the Arctic, *Department of Earth Sciences, University of Southern California, Los Angeles, CA*, [Invited Seminar]

2015

29. **McKay, N.P.** (2015a), Collaborative database building and lessons learned from the PAGES 2k projects, *Open kickoff meeting of the PAGES working group Global Soil and Sediment Transfers in the Anthropocene, Hamburg, Germany*, [Invited talk]
28. **McKay, N.P.** (2015b), The State (and Future) of Age Modelling, *Proxy System Modeling Workshop, Catalina Island, CA*, [Invited Talk]
27. **McKay, N.P.** (2015c), GeoChronR and ChronRater, *Cyber for Paleo Webinar Series*, [Invited Talk]
26. **McKay, N.P.**, D. S. Kaufman, P. D. Zander, and C. C. Routson (2015a), The timing and amplitude of Holocene Neoglaciation in the Arctic, *AGU Fall Meeting 2015, San Francisco, California*, [Talk]
25. **McKay, N.P.**, J. Emile-Geay, and the PAGES2k Consortium (2015b), A global multiproxy database for temperature reconstructions of the Common Era, *AGU Fall Meeting 2015, San Francisco, California*, [Invited Talk]
24. **McKay, N.P.**, D. S. Kaufman, P. D. Zander, and C. C. Routson (2015c), The timing and amplitude of Holocene Neoglaciation in the Arctic, *Department of Geosciences, University of Arizona, Tucson, Arizona*, [Invited Seminar]

2014

23. **McKay, N.P.** (2014b), PAGES-Powell North America 2k Database, *AGU Fall Meeting 2014, San Francisco, California*, [Poster]
22. **McKay, N.P.**, and D. Kaufman (2014b), The spatial and temporal structure of the onset of Neoglaciation in the Arctic, *PAGES Holocene Workshop, Mt. Hood, OR*, [Invited Talk]
21. **McKay, N.P.** (2014c), The more daunting aspects of the (paleo)data, *SAMSI paleoclimate seminar series*, [Invited Talk]
20. **McKay, N.P.** (2014d), Advances in climate field reconstructions and North America 2k, *LOTRED-SA 3rd International Symposium, Medellin, Columbia*, [Invited Talk]
19. **McKay, N.P.**, D. Kaufman, H. S. S. Sundqvist, and N. L. Balascio (2014), The Arctic Holocene Transitions database and ChronRater: a simple approach to assessing the accuracy of age models from Holocene sediment cores, *Age Models, Chronologies and Databases workshop, Belfast, UK*, [talk]
18. **McKay, N.P.** (2014e), Development of the global PAGES 2k temperature database, *PAGES2k AGU 2014 Satellite Meeting*, [Invited Talk]

2013

17. **McKay, N.P.**, and D. Kaufman (2013), North-south contrast in climate change across northwestern North America during the mid-Holocen transition and the possible relation to western Arctic Ocean sea ice., *AGU Fall Meeting 2013, San Francisco, CA*, [Poster]
16. **McKay, N.P.** (2013), Site-level analyses in PAGES2k, *PageS2k-PMIP3 Workshop, Madrid, Spain*, [Invited Talk]
15. **McKay, N.P.**, H. S. Sundqvist, D. S. Kaufman, and N. L. Balascio (2013b), On the power of a well-formed database: A proxy climate database for the mid-Holocene transition in the Arctic, *3rd International Climate Informatics Workshop, Boulder, Colorado*, [Spotlight talk and poster]
14. **McKay, N.P.**, J. T. Overpeck, T. M. Shanahan, E. Brown, J. A. Peck, C. W. Heil, J. W. King, and C. A. Scholz (2013c), A high-resolution record of West African Monsoon variability for the past 530,000 years from Lake Bosumtwi, southern Ghana, *4th PAGES Open Science Meeting, Goa, India*, [talk]

2013

13. **McKay, N.P.**, J. T. Overpeck, T. M. Shanahan, J. A. Peck, C. W. Heil, J. W. King, and C. A. Scholz (2013d), A 12,000-Year-Long, Annually-Resolved Varve Record Spanning the Last Interglacial from Lake Bosumtwi, Southern Ghana, *2nd PAGES Young Scientist's Meeting, Goa, India*, [Poster]

2012

12. **McKay, N.P.**, and D. S. Kaufman (2012), Data-model comparison of the Mid-Holocene Transition in the Arctic, *2012 PAGES SynTraCE-21 Workshop, Providence, Rhode Island*, [talk]
11. **McKay, N.P.** (2012), Pleistocene climate variability in tropical West Africa on interannual to orbital timescales, inferred from the Lake Bosumtwi sediment record, *School of Earth Sciences and Environmental Sustainability Seminar Series, Fall 2012, Northern Arizona University, Flagstaff, Arizona*, [talk]
10. **McKay, N.P.**, T. R. Ault, and S. St. George (2012a), Improving access to the global paleoclimate dataset through the development of a virtual paleoclimate laboratory in R (vplR), *2nd International Climate Informatics Workshop, Boulder, Colorado*, [Spotlight talk and poster]
9. **McKay, N.P.**, J. T. Overpeck, T. M. Shanahan, E. T. Brown, J. A. Peck, J. W. King, C. Scholz, and C. W. Heil (2012b), Paleohydrologic changes in tropical West Africa on interannual to orbital timescales over the past 500,000 years, *Colloquium of the French Academy of Science on the African Tropical Forest, Paris, France*, [Invited Talk]

2011

8. **McKay, N.P.**, J. T. Overpeck, T. M. Shanahan, J. A. Peck, C. W. Heil, J. W. King, and C. A. Scholz (2011b), A 12,000-Year-Long, Annually-Resolved Varve Record Spanning the Last Interglacial from Lake Bosumtwi, Southern Ghana, *AGU Fall Meeting 2012, San Francisco, California*, [talk]
7. **McKay, N.P.**, J. T. Overpeck, T. M. Shanahan, J. A. Peck, J. W. King, C. A. Scholz, and C. W. Heil (2011c), Interannual- to multicentiennial-scale variability in the West African Monsoon during the Eemian, *AGU Fall Meeting 2011, San Francisco, California*, [Poster]
6. **McKay, N.P.**, J. T. Overpeck, and B. Otto-Bliesner (2011d), The role of ocean thermal expansion in Last Interglacial sea level rise, *Geodaze 2011, Tucson, Arizona*, [talk]
5. **McKay, N.P.**, J. T. Overpeck, T. M. Shanahan, J. A. Peck, J. W. King, C. A. Scholz, and C. W. Heil (2011e), An initial comparison of Eemian and Holocene varve structure and variability in the Lake Bosumtwi drill core record, *2nd Workshop of PAGES Varves Working Group, Corpus Christi, Texas*, [talk]

2010

4. **McKay, N.P.**, J. T. Overpeck, E. T. Brown, T. M. Shanahan, J. A. Peck, J. W. King, Scholz, C. A., and C. W. Heil (2010), A scanning-XRF record of Lake Bosumtwi sediments: Implications for West African Monsoon variability over the past 500 kyr, *Geodaze 2010, Tucson, Arizona*, [talk]

2009

3. **McKay, N.P.**, J. T. Overpeck, E. T. Brown, T. M. Shanahan, J. A. Peck, J. W. King, Scholz, C. A., and C. W. Heil (2009), A scanning-XRF record of Lake Bosumtwi sediments: Implications for West African Monsoon variability over the past 500 kyr, *AGU Fall Meeting 2009, San Francisco, California*, [talk]

2007

2. **McKay, N.P.**, and D. S. Kaufman (2007), A biogenic-silica-inferred temperature reconstruction for the past 2 kyr, and Holocene climate fluctuations from Hallet and Greyling Lakes, Chugach Range, south-central Alaska, *International Arctic Workshop, Skaftafell, Iceland*, [talk]

2005

1. **McKay, N.P.**, D. S. Kaufman, and D. E. Anderson (2005), Characterization of climatic influences on modern sedimentation in an arctic lake, Svalbard, Norway, *Northeast regional Geological Society of America Conference, Saratoga Springs, New York*, [talk]

Research Grants & Proposals

Since 2013: A total of twenty four externally funded projects for $7.1M of local (NAU) funding from the National Science Foundation, Belmont Forum, US Geological Survey, New Zealand Ministry of Business, Innovation and Employment and private funding. Overall funding rate of 46%; 75% as Project Lead PI.

2026 Collaborative Research: CAIG: Advancing Holocene paleoclimate reconstruction and physics-informed neural networks through lake model emulation – [**NAU Lead Investigator**] – *Funded*
- *NSF: CAIG*
- NSF award 2615988 (08/2026 to 07/2029)
- Requested funds (local): $485,329

2025 Collaborative Research: Innovative Resources: GENESIS: GENerating Earth Science Infrastructure from Seminal studies – [**Co-Investigator**] – *Declined*
- *NSF: Geoinformatics*
- Requested funds (local): $492,804

2025 Collaborative Research: GEO-OSE: Track 2: Building an open science community and services for biomarker researchers in the geosciences – [**NAU Lead Investigator**] – *Funded*
- *NSF: OSE*
- NSF award 2606091 (10/2026 to 09/2029)
- Requested funds (local): $79,895

2025 Collaborative Research: GEO-OSE: Track 1: Building an open science community and shared reproducible data flow for lipid biomarker researchers – [**NAU Lead Investigator**] – *Declined*
- *NSF: OSE*
- Requested funds (local): $36,590

2024 Collaborative Research: Assessing the ecoclimate sensitivity of North American biome-scale ecotones to climate changes since the last ice age – [**NAU Lead Investigator**] – *Funded*
- *NSF: DEB*
- NSF award 2516454 (09/2025 to 08/2028)
- Requested funds (local): $457,275

2024 Investigating spatial patterns of temperature and precipitation in the PATCH project – [**Co-Investigator**] – *Declined*
- *NSF: CLD*
- Requested funds (local): $599,314

2024 Collaborative Research: A Synthesis of Hydroclimate Records from Lake Bosumtwi and other African Scientific Drill Cores: New Proxies, Geochronologies and Proxy-Model Integration – [**NAU Lead Investigator**] – *Declined*
- *NSF: P4Climate*
- Requested funds (local): $114,429

2024 Sustained Resources: A platform to support reproducible paleogeoscience: facilitating data access, analysis, and visualization with PReSto – [**Project Lead Investigator**] – *Declined*
- *NSF: Geoinformatics*
- Requested funds (local): $499,447

2024 Collaborative Research: Paleo Records Of GLacier And Climate changes Inferred from Alaskan Lakes (PROGLACIAL), supplement – [**Investigator**] – *Funded*
- *NSF: P4Climate*
- Added to NSF award 2303462 in 06/2024
- Requested funds (local): $86,560

2024 Collaborative Research: Elements: TUPS: Table Understanding for Paleoclimate Studies – [**NAU Lead Investigator**] – *Funded*
- *NSF: CSSI*
- NSF award 2411268 (09/2024 to 08/2027)
- Requested funds (local): $171,570

2023 Collaborative Research: Patterns of African Hydroclimate: New Geochronologies and Proxy-Model Integration of Million-year Legacy Drill Core Records from Lakes Bosumtwi and Malawi – [**NAU Lead Investigator**] – *Declined*
- *NSF: P4Climate*
- Requested funds (local): $101,664

2023 Collaborative Research: GEO OSE Track 2: FORAES3: An integrative Framework for Open, Reproducible and Agile Earth System Science Synthesis – [**Project Lead Investigator**] – *Declined*
- *NSF: GEO Open Science Ecosystem*
- Requested funds (local): $574,090

2023 The Arctic permafrost carbon conundrum: will warmer temperatures lead to enhanced Arctic soil carbon emissions or enhanced soil carbon storage over long timescales? – [**Investigator**] – *Declined*
- *NSF: Arctic System Science*
- Requested funds (local): $1,097,472

2023 Collaborative Research: GEO OSE Track 1: FROGS: Facilitating Reproducible Open GeoScience – [**NAU Lead Investigator**] – *Funded*
- *NSF: GEO Open Science Ecosystem*
- NSF award 2324733 (01/2024 to 12/2026)
- Requested funds (local): $84,520

2023 Testing amino acid paleothermometry in radiocarbon-dated lake sediment– [**Investigator**] – *Funded*
- *NSF: GG*
- NSF award 2317409 (08/2023 to 06/2027)
- Requested funds (local): $600,527

2022 Collaborative Research: Paleo Records Of GLacier And Climate changes Inferred from Alaskan Lakes (PROGLACIAL) – [**Investigator**] – *Funded*
- *NSF: P4Climate*
- NSF award 2303462 (06/2023 to 05/2027)
- Requested funds (local): $496,646

2021 Collaborative Research: GCR: Linking preindustrial with modern climate indicators to inform future climate change – [**Investigator**] – *Declined*
- *NSF: Growing Convergent Research*
- Requested funds (local): $1,264,647

2021 Data synthesis and analysis of carbon climate dynamics on millennial timescales: Implications for 21st Century climate – [**Investigator**] – *Declined*
- *NSF: Arctic System Science*
- Requested funds (local): $1,481,022

2021 EarthCube Capabilities: PaleoCube: Enabling Cloud-Based Paleoclimatology– [**NAU Lead Investigator**] – *Funded*
- *NSF: Earthcube*
- NSF award 2126510 to USC (09/2021 to 08/2027)
- Requested funds (local): $105,581

2020 Collaborative Research: EarthCube Data Capabilities: PaleoCube: next-generation paleoscience with EarthCube – [**NAU Lead Investigator**] – *Declined*
- *NSF: Earthcube*
- Requested funds (local): $251,028

2020 Dust cycle monitoring in the 4 corners region – [**NAU Lead Investigator**] – *Funded*
- *Bob and Judi Braudy Foundation*
- Requested funds (local): $80,000

2020 Collaborative Research: AccelNet: LINKEDECOS: A Linked Ecosystem of Open Data Resources and Scientists to Investigate Past Biodiversity Dynamics in Socio-environmental Systems – [**NAU Lead Investigator**] – *Declined*
- *NSF: Accelnet*
- Requested funds (local): $196,908

2020 Monsoon 21k: A global monsoon synthesis and data model comparison – [**Investigator**] – *Funded*
- *NSF: Paleo-perspectives on climate change*
- NSF award 2002328 (09/2020 to 08/2026)
- Requested funds (local): $443,211

2020 A machine readable paleoseismic event database: toward improving long-term earthquake predictability in southern California – [**Investigator**] – *Funded*
- *NSF: Southern California Earthquake Center*
- Requested funds (local): $32,782

2020 Collaborative Research: PReSto: A Paleoclimate Reconstruction Storehouse to Broaden Access and Accelerate Scientific Inference – [**Project Lead Investigator**] – *Funded*
- *NSF: Geoinformatics*
- NSF award 1948746 (07/2020 to 06/2026)
- Requested funds (local): $453,467

2020 Collaborative Research: Patterns and processes of abrupt Arctic warming based on paleoclimate observations and models – [**NAU Lead Investigator**] – *Funded*
- *NSF: ARCSS*
- NSF award 1948005 (07/2020 to 06/2026)
- Requested funds (local): $527,906

2019 EarthCube Data Capabilities: Collaborative Proposal: PaleoWorkBench - enabling next-generation Earth system science – [**NAU Lead Investigator**] – *Declined*
- *NSF: Earthcube*
- Requested funds (local): $247,585

2019 EarthCube Data Capabilities: Collaborative Proposal: Reducing Time-To-Science in the Earth Sciences: Annotations to foster convergence, inclusion, and credit – [**NAU Lead Investigator**] – *Funded*
- *NSF: Earthcube*
- NSF award 1928320 (09/2019 to 08/2023)
- Requested funds (local): $209,116

2019 Collaborative Research: AccelNet: LINKEDECOS: A Linked Ecosystem of Open Data Resources and Scientists to Investigate Past Biodiversity Dynamics in Socio-environmental Systems – [**NAU Lead Investigator**] – *Declined*
- *NSF: Accelnet*
- Requested funds (local): $210,710

2019 MRI: Acquisition of equipment for an integrated gas analysis and labeling radiocarbon system with a focus on Arctic carbon and geochronology – [**Investigator**] – *Funded*
- *NSF: MRI*
- NSF award 1919506 (08/2019 to 07/2023)
- Requested funds (local): $486,652

2018 Collaborative Research: Quantifying Holocene climate variations through data assimilation using proxies and GCM output – [**Investigator**] – *Funded*
- *NSF: Paleo-perspectives on climate change*
- NSF award 1903548 (09/2019 to 08/2025)
- Requested funds (local): $441,947

Abrupt Change in Climate and Ecosystems: Where are the Tipping Points? [**Project Lead Investigator**] – *Funded*
- *Belmont Forum: Science-driven e-Infrastructure Innovation*
- NSF award 1929460 (07/2019 to 06/2023)
- Requested funds (local): $152,451

2018 Collaborative Research: Framework: Software: Community Earth System Informatics: Enabling Convergent Science [**NAU Lead Investigator**] – *Declined*
- *NSF: Cyberinfrastructure for Sustained Scientific Innovation*
- Requested funds (local): $498,835

2018 MRI: Acquisition of automated graphitization equipment for radiocarbon analysis: Arctic carbon, geochronology, and other applications – [**Investigator**] – *Declined*
- *NSF: Major Research Instrumentation*
- Requested funds (local): $411,735

2017 Collaborative Research: Quantifying Holocene climate variations through data assimilation using proxies and GCM output – [**Investigator**] – *Declined*
- *NSF: Paleo-perspectives on climate change*
- Requested funds (local): $443,129

2017 Collaborative Research: Evolution of terrestrial climate and vegetation through the mid-Pleistocene climate transition: A 1.5 Ma record from Stoneman Lake, Arizona– [**Investigator**] – *Declined*
- *NSF: Paleo-perspectives on climate change*
- Requested funds (local): $323,472

2017 Holocene Dustiness in the San Juan Mountains, Colorado – [**Investigator**] – *Declined*
- *NSF: Paleo-perspectives on climate change*
- Requested funds (local): $451,234

2017 Lakes380: past, present, future: Our lakes' health: past, present, future - Me hoki whakamuri kia haere whakamua – [**NAU Lead Investigator**] – *Funded*
- *New Zealand Ministry of Business, Innovation and Employment – Endeavour Fund*
- Local funds: $92,418

2017 Collaborative Proposal: EarthCube Integration: THROUGHPUT: Standards and Services for Community Curated Repositories – [**NAU Lead Investigator**] – *Funded at Pilot Level*
- *NSF: Earthcube*
- NSF award 1740667 (09/2017 to 08/2019)
- Local funds (pilot): $69,277

2017 MRI: Acquisition of A Mini Carbon Dating System for Understanding Arctic Carbon and Climate (ACCLIMATE) – [**Investigator**] – *Declined*
- *NSF: Major Research Instrumentation*
- Requested funds (local): $1,982,320

2016 MRI: Acquisition of A Mini Carbon Dating System for Understanding Arctic Carbon and Climate (ACCLIMATE) – [**Investigator**] – *Declined*
- *NSF: Major Research Instrumentation*
- Requested funds (local): $1,971,874

2016 Synthesis of Holocene Proxy Climate Records from Western North America – [**Investigator**] – *Funded*
- *NSF: Paleo-perspectives on climate change*
- NSF award 1602105 (09/2016 to 08/2020)
- Local funds: $482,388 (2019 supplement increased to $522,048)

2016 Collaborative Research: Evolution of terrestrial climate and vegetation through the mid-Pleistocene climate transition: A 1.2 Ma record from Stoneman Lake, Arizona – **[Investigator]** – *Declined*
- *NSF: Paleo-perspectives on climate change*
- Requested funds (local): $399,774

2016 Holocene Dustiness in the San Juan Mountains, Colorado – **[Investigator]** – *Declined*
- *NSF: Paleo-perspectives on climate change*
- Requested funds (local): $274,000

2016 The Drought-Dust Nexus in the Four-Corners Region and Impacts on Society – **[Lead Investigator]** – *Funded*
- *Bob and Judi Braudy Foundation*
- Local funds: $139,308

2015 EarthCube IA: Collaborative Proposal: LinkedEarth: Crowdsourcing Data Curation & Standards Development in Paleoclimatology **[NAU Lead Investigator]** – *Funded*
- *NSF: Earthcube: Integrated Activities*
- NSF award 1540996 (09/2015 to 08/2018)
- Local funds: $113,015

2015 EarthCube IA: Collaborative Proposal: Chains of Provenance: Linking geoscientific archives, repositories, and analysis **[Investigator]** – *Declined*
- *NSF: Earthcube: Integrated Activities*
- Requested funds (local): $132,515

2015 Collaborative Research: Evolution of terrestrial climate and vegetation through the mid-Pleistocene climate transition: A 1.2 Ma record from Stoneman Lake, Arizona – **[Investigator]** – *Declined*
- *NSF: Paleo-perspectives on climate change*
- Requested funds (local): $320,589

2015 Holocene Dustiness in the San Juan Mountains, Colorado – **[Investigator]** – *Declined*
- *NSF: Paleo-perspectives on climate change*
- Requested funds (local): $203,595

2014 Collaborative research: Developing a system model of Arctic glacial-lacustrine sedimentation for investigating past and future climate change **[Lead Investigator]** – *Funded*
- *NSF: Arctic System Science*
- NSF award 1418000 (01/2015 to 12/2018)
- Local Funds: $753,204 plus $300,000 logistical support

2014 Collaborative Research: GeoChronR – Open-source Tools for the Analysis, Visualization and Integration of Time-Uncertain Geoscientific Data **[Lead Investigator]** – *Funded*
- *NSF: Geoinformatics*
- NSF award 1347221 (07/2014 to 06/2018)
- Local funds: $369,348

2013 EarthCube Building Blocks: Crowdsourcing Geoscientific Data Curation **[NAU Lead Investigator]** – *Declined*
- *NSF: Earthcube Building Blocks*
- Requested funds (local): $66,179

2013 A broader view of North American climate over the past two millennia: Synthesizing paleoclimate records from diverse archives [**Lead Investigator**] – *Funded*
- *USGS: Powell Center*
- Local funds: $129,737

2012 Collaborative Research: A virtual paleoclimate laboratory to quantify climate reconstruction uncertainty – [**Lead Investigator**] – *Declined*
- *NSF: Paleo-perspectives on climate change*
- Requested funds (local): $473,326

2012 Collaborative Research (RUI): Alaskan and Peruvian Contributions to a Synthesis of Holocene Proxy Climate Records from the Arctic and Eastern Pacific Rim [**Investigator**] – *Declined*
- *NSF: Paleo-perspectives on climate change*
- Requested funds (local): $687,900

2011 NSF Postdoctoral Research Fellowship – [**PI**] – *Declined*
- *NSF: Postdoctoral Research Fellowship*
- Requested funds: $149,875

Teaching

2026	**Instructor**	Climate Change (ENV 115), Earth and Environmental Data Analysis (EES 680), Northern Arizona University
2025	**Instructor**	Atmosphere and Hydrosphere (ENV 360), Climate Change (ENV 115), Northern Arizona University
2024	**Instructor**	Atmosphere and Hydrosphere (ENV 360), Earth and Environmental Data Analysis (EES 680), R Package Development Seminar (EES 698), Northern Arizona University
2023	**Instructor**	Climate Change (ENV 115), Earth and Environmental Data Analysis (EES 680), 4.2 ka event seminar (EES 698), Northern Arizona University
2022	**Instructor**	Climate Change (ENV 115), Earth and Environmental Data Analysis (EES 680), Northern Arizona University
2021	**Instructor**	Climate Change (ENV 115), Earth and Environmental Data Analysis (EES 680), Northern Arizona University
2019	**Instructor**	Climate Change (ENV 115), Climate Change Methods (ENV 199),Earth and Environmental Data Analysis (EES 680), Northern Arizona University
2018	**Instructor**	Climate Change (ENV 115), Global Climate Change (ENV 595), Earth and Environmental Data Analysis (EES 680), Climate Dynamics (EES 580), Northern Arizona University
2017	**Instructor**	Climate Change (ENV 115), Earth and Environmental Data Analysis (EES 599), Northern Arizona University
2016	**Instructor**	Climate Change (ENV 115), Atmosphere & Hydrosphere (ENV 360), Earth and Environmental Data Analysis (EES 599), Climate Dynamics (EES 599), Paleoclimate Seminar (EES 698), Northern Arizona University
2015	**Instructor**	Climate Change (ENV 115), Climate Dynamics (EES 599), Northern Arizona University
2014	**Instructor**	Climate Change (ENV 115), Paleolakes seminar (EES 698), Northern Arizona University
2013	**Instructor**	Modes of Climate Variability (EES 698), Northern Arizona University

Awards

2022	Eric Grimm Award for Excellence in Open Science and Data Stewardship
2021	Fulbright New Zealand Scholar
2018	AGU 2017 Editors Citation for Excellence in Refereeing
2016	Northern Arizona University Research and Creativity Award: "Most Promising New Scholar"
2012	William G. McGinnies Award for Arid Lands Research

Professional Service and Outreach

2026	Leader: PReSto Project
	Co-Leader: LinkedEarth
	Lead organizer: PReSto workshop (June)

	Data Coordinator: Hydroclimate2k
	NSF ad hoc proposal reviewer
	Referee: *Geoscientific Model Development*
	Senator (CEFNS), NAU Faculty Senate
2025	Leader: PReSto Project
	Co-Leader: LinkedEarth
	Co-organizer: FROGS workshop (June)
	Data Coordinator: Hydroclimate2k
	Member of FAIROS RCN Advisory Council
2024	Leader: ACCEDE and PReSto Projects
	Co-Leader: LinkedEarth
	Co-organizer: PyRates workshop (June)
	Member of FAIROS RCN Advisory Council
2023	Leader: ACCEDE and PReSto Projects
	Co-Leader: LinkedEarth
	Co-organizer: Bring-your-own-data LinkedEarth Hackathon (Jun)
	NSF Panelist
	Member of FAIROS RCN Advisory Council
2022	Leader: ACCEDE and PReSto Projects
	Co-Leader: LinkedEarth
	Organizer: geoChronR Hackathon (Feb)
2021	Leader: ACCEDE and PReSto Projects
	Co-Leader: LinkedEarth
	Data coordinator: iso2k project
	Instructor: Pyleoclim Hackathons (Feb & Oct)
2020	Leader: ACCEDE and PReSto Projects
	Co-Leader: LinkedEarth
	Data coordinator: Temperature 12k, iso2k projects
	NOAA Climate Observation and Monitoring program review panelist
2019	Leader: ACCEDE Project
	Co-Leader: LinkedEarth
	Data coordinator: Temperature 12k, iso2k projects
2018	Co-Leader: LinkedEarth
	Data coordinator: iso2k project
2017	Leader: PAGES North America 2k working group
	Organizer: 2nd geoChronR workshop, Flagstaff, AZ
	Co-Leader: LinkedEarth
	Data coordinator: Global PAGES 2k Network; iso2k project
2016	Leader: PAGES North America 2k working group
	Organizer: 1st geoChronR workshop, Flagstaff, AZ
	Co-Leader: LinkedEarth
	Data coordinator: Global PAGES 2k Network; iso2k project
2015	Leader: PAGES North America 2k working group

	Data coordinator: Global PAGES 2k Network; iso2k project
	Organizer : PAGES/USGS Powell Center "North America 2k working group meeting", 12-15 October Fort Collins, CO
2014	Leader: PAGES North America 2k working group
	Data coordinator: Global PAGES 2k Network
	Organizer : PAGES/USGS Powell Center "North America 2k working group meeting", 23-26 June Fort Collins, CO
	Organizer : PAGES "North America 2k AGU Breakout meeting", 17 December, San Francisco, CA
	Co-organizer: "PAGES 2k - Advances in Climate Field Reconstruction Workshop", 15-16 April, Woods Hole, MA
	Invited panelist for NOAA Paleoclimate ad-hoc review panel
2013	Climate science Q&A accompanying *Chasing Ice*
2012	Geology display at Knoles Elementary School STEM Night
	Public Lecture – Stories in the mud: What lakes have taught me about past climates. Inaugural McGinnies Lecture, University of Arizona.

Reviewer (Journals)

Since 2011 I have peer-reviewed dozens of manuscripts for the following journals: *Science; Nature; Nature Geoscience; Nature Communications; Nature Communications: Earth & Environment; Geophysical Research Letters; Climate of the Past, JGR: Biogeosciences; Paleoceanography and Paleoclimate; Earth-Science Reviews; Scientific Data; JGR: Biogeoscience; Journal of Paleolimnology; Quaternary Science Reviews; Palaeogeography, Palaeoclimatology, Palaeoecology; JGR: Oceans; Geochemical Transactions; Geoscientific Model Development.*

I also received the AGU 2017 Editors Citation for Excellence in Refereeing for my reviews of a paper in *Geophysical Research Letters.*

Media Coverage

Print	Arizona Republic, Arizona Daily Star, BBC News, Green Valley News and Sun, NAU Lumberjack, NAU News, NBC News, and Science Daily
Radio	BBC
Television	Arizona Public Media, AZ Family News (Channel 3)
New Media	Nature Publishing Group Journal Club, PAGES 2k Google Hangout

Students and Postdocs Supervised

Served as lead advisor

Postdocs	Michael Erb (2017-2019), Allison Cluett (2021-2022)
Ph.D. Students	Stéphanie Arcusa (2016-2020), Chris Hancock (2020-2024), Ethan Yackulic (2020-2026), Frank Telles (2021-2026)
M.S. Students	Eli Allred (2024 - Present), Sela Patterson (2019-2021), Charlotte Wiman (2017-2019), Ethan Yackulic (2015-2017)

Served on committee

Ph.D. Students	Ellie Broadman, Stacy Kish, Emily Romano, Leah Marshall, Emma Lathrop, Beatrice Bock, Brandon McNamara, Anna Gagne-Landmann, Hunter Allen, Anastasia Pulak

M.S. Students Annie Wong, Lindsey Gipson, Rebecca Ellerbroek, Charles Mogen, Chris Benson, Rachel Krueger, Lorna Thurston, Hannah Kolus, Doug Steen, Paul Zander, Scarlett Hunt, Caitlin Walker, Bri McMaster Smith, Ryan Ellis